

Localized surface plasmons in layered spherical systems with $\epsilon_r \neq \epsilon_t$ with account for surface scattering

In this set of notes, we address the consequences of going beyond two of the typical assumptions of nano-plasmonics, as detailed below

In **Section 1**, **Section 2**, **Section 3**, and **Section 4**, we examine nanoshells and nanospheres composed of radially anisotropic materials. By radially anisotropic we mean that the permittivity tensor, ϵ , is diagonal in the spatially varying spherical coordinate basis. That is,

$$\epsilon = \epsilon_r \mathbf{e}_r \mathbf{e}_r + \epsilon_\theta \mathbf{e}_\theta \mathbf{e}_\theta + \epsilon_\phi \mathbf{e}_\phi \mathbf{e}_\phi,$$

where $\epsilon_r \neq \epsilon_\theta = \epsilon_\phi$. The papers we follow (or that will be useful) in these four sections are

Influence of spherically anisotropic core on the optical properties of gold nanoshell

(<https://link.springer.com/content/pdf/10.1007/s00339-008-4951-y.pdf>) by **Wu and Liu (Paper 1)**, Thin-shell

approximation of Mie theory for a thin anisotropic layer spaced away from a spherical core:

Application to dye-coated nanostructures

(<https://journals.aps.org/pr/abstract/10.1103/PhysRevA.104.033502>) by **Tang et al (Paper 2)**, Scattering by rotationally symmetric anisotropic spheres: Potential formulation and parametric studies

(<https://journals.aps.org/pre/pdf/10.1103/PhysRevE.75.026609>) by **Qiu et al (Paper 3)** and Anomalous

absorption, plasmonic resonances, and invisibility of radially anisotropic spheres

(<https://ieeexplore.ieee.org/abstract/document/7771802>) by **Wallen et al (Paper 4)**,

Electromagnetic theory of tunable SERS manipulated with spherical anisotropy in coated

nanoparticles (https://www.ece.nus.edu.sg/stfpage/eleqc/ACS_JPC.pdf) by **Yin et al (Paper 5)**, Mie

scattering from thin anisotropic spherical shells (https://pubs.aip.org/aip/jcp/article-pdf/92/8/4643/18986170/4643_1_online.pdf) by **Lange and Aragon (Paper 6)**, and Electromagnetic

transparency by coated spheres with radial anisotropy

(<https://journals.aps.org/pre/pdf/10.1103/PhysRevE.78.046609>) by **Gao et al (Paper 7)**.

Section 1, **Section 2** and **Section 3** are all focused on the quasistatic limit. Thus, in those sections we assume the electric field may be well described through an electrostatic potential. In **Section 4**, we use a full-wave approach to find the form of the electromagnetic fields when retardation is taken into account exactly.

In **Section 5**, we develop a more sophisticated calculation for surface scattering in a small sphere using a hybrid transport theory/geometrical approach. In so doing, we leverage some of the approximations made in the paper The electrical conductivity of thin wires ([https://www.jstor.org/stable/98505?](https://www.jstor.org/stable/98505?casa_token=mUYsRexsxlAAAAA:8_TPk0vwFKg_hkbG_o74JwC9cTcRvzK9yHoduHT8j6FOQQN1OQ65BATO9q_A-fLzprjTSR_AT9UuM4DLqP3PzJ_VSsFySmKjCHaCV2vJ3ryLzuk9au8)
[casa_token=mUYsRexsxlAAAAA:8_TPk0vwFKg_hkbG_o74JwC9cTcRvzK9yHoduHT8j6FOQQN1OQ65BATO9q_A-fLzprjTSR_AT9UuM4DLqP3PzJ_VSsFySmKjCHaCV2vJ3ryLzuk9au8](https://www.jstor.org/stable/98505?casa_token=mUYsRexsxlAAAAA:8_TPk0vwFKg_hkbG_o74JwC9cTcRvzK9yHoduHT8j6FOQQN1OQ65BATO9q_A-fLzprjTSR_AT9UuM4DLqP3PzJ_VSsFySmKjCHaCV2vJ3ryLzuk9au8)). (**Paper 12**) by **R.B Dingle**. In **Section 6**, we verify the results of the paper Supplementary Information: Ultralow Electron-Surface Scattering in Nanoscale Metals Leveraging Fermi Surface Anisotropy (<https://journals.aps.org/prmaterials/supplemental/10.1103/PhysRevMaterials.6.085002/Supp.pdf>) by **Kumar et al** (**Paper 8**) for surface scattering calculations in a thin rectangular wire. In **Section 7**, we move to a more complicated geometry, i.e. a small metallic sphere (addressed with a full transport theory calculation). In this section, we verify some of the results reported in Kinetic theory of electroconductivity of metal nanoparticles in the condition of surface plasmon resonance (<https://www.sciencedirect.com/science/article/pii/S2666523921000039#bib0010>) by **Semchuk et al** (**Paper 9**), Optical and transport properties of spheroidal metal nanoparticles with account for the surface effect ([https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.085448?](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.085448?casa_token=fDOIGdDfpDMAAAAA%3A_AJIEdR4_vkqK5WLC9Dj9v7neUMsP6SonZfU6vyJ_wkL_pvvRzXtH-DeB6aT2Fcy0glzm2UCjZitZvc)
[casa_token=fDOIGdDfpDMAAAAA%3A_AJIEdR4_vkqK5WLC9Dj9v7neUMsP6SonZfU6vyJ_wkL_pvvRzXtH-DeB6aT2Fcy0glzm2UCjZitZvc](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.085448?casa_token=fDOIGdDfpDMAAAAA%3A_AJIEdR4_vkqK5WLC9Dj9v7neUMsP6SonZfU6vyJ_wkL_pvvRzXtH-DeB6aT2Fcy0glzm2UCjZitZvc)) by **Grigorchuk and Tomchuk** (**Paper 10**), Shape and size effects on the energy absorption by small metallic particles (<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.73.155423>) by **Tomchuk and Grigorchuk** (**Paper 11**).

Section 1: Resonance solution in the quasistatic approximation

In this section, we find the allowed functional forms of the electric potential in a dielectric with radial anisotropy. In the quasistatic approximation, we have $\mathbf{E}(\mathbf{r}, \omega) = -\nabla\phi(\mathbf{r}, \omega)$, which means that the electrostatic potential, $\phi(\mathbf{r}, \omega)$, must satisfy the following differential equation in a region with no free charges:

$$\nabla \cdot [\epsilon \nabla \phi(\mathbf{r}, \omega)] = 0 \rightarrow (\epsilon_r(\omega)/\epsilon_t(\omega)) \partial_r \left[r^2 \partial_r \phi(r, \theta, \omega) \right] + \frac{1}{\sin(\theta)} \partial_\theta \left[\sin(\theta) \partial_\theta \phi(r, \theta, \omega) \right] = 0$$

In writing this equation, we have assumed the solution to be azimuthally symmetric, for convenience. By separating variables and demanding $\phi(r, \theta, \omega) = P_l(\cos(\theta)) f(r, \omega)$, we obtain the following simple differential equation in the radial coordinate:

$$\frac{\epsilon_r(\omega)}{\epsilon_t(\omega)} \partial_r \left[r^2 \partial_r f(r, \omega) \right] = l(l+1) f(r, \omega),$$

where we used the following (**Legendre's equation**):

$$\frac{1}{\sin(\theta)} \partial_{\theta} \left[\sin(\theta) \partial_{\theta} P_l(\cos(\theta)) \right] = \partial_x \left[(1 - x^2) \partial_x P_l(x) \right]_{x=\cos(\theta)} = -l(l+1) P_l(\cos(\theta))$$

Thus, if we take a polynomial solution in the radial coordinate, $f(r, \omega) = r^{\alpha(\omega)}$, we obtain the following equation for the exponent, $\alpha(\omega)$:

$$\begin{aligned} \frac{\varepsilon_r(\omega)}{\varepsilon_t(\omega)} \alpha(\omega)(\alpha(\omega) + 1) &= l(l+1) \rightarrow \\ \alpha(\omega) &= \frac{-\varepsilon_r(\omega)/\varepsilon_t(\omega) \pm \sqrt{(\varepsilon_r(\omega)/\varepsilon_t(\omega))^2 + 4(\varepsilon_r(\omega)/\varepsilon_t(\omega))l(l+1)}}{2\varepsilon_r(\omega)/\varepsilon_t(\omega)} = \\ &= -\frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{l(l+1)\varepsilon_t(\omega)}{\varepsilon_r(\omega)}}, \end{aligned}$$

which is consistent with the discussion in **Paper 1** (top paragraph in the second column on the second page) for the dipolar, $l = 1$, channel. In addition, this equation is identical to **Equation 6** of **Paper 4** and **Equation 17** of **Paper 2**. Lastly, we note the following useful relation:

$$\nu \equiv -\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{l(l+1)\varepsilon_t(\omega)}{\varepsilon_r(\omega)}} = -1 - \left[-\frac{1}{2} - \sqrt{\frac{1}{4} + \frac{l(l+1)\varepsilon_t(\omega)}{\varepsilon_r(\omega)}} \right]$$

Thus, we may write the potential polynomial solutions as $r^{\nu}, r^{-\nu-1}$, consistent with **Equation 3** of **Paper 4**.

Section 2: Electric field profile for a nanoshell/anisotropic core system in a uniform electric field

From **Section 1**, we know the general spatial profile of the electric field in a radially anisotropic material. What remains is to use Maxwell boundary conditions to determine the response to an applied external field (or find the plasmonic solutions in the absence of an applied field). In view of this goal, in this section, we wish to prove **Equation 6**, **Equation 7** and **Equation 8** of **Paper 1**. These give us the field profiles everywhere in space given an applied field, $\mathbf{E}_0 = E_0 \mathbf{e}_z$. Note that from the arguments in the previous section, we have already proven **Equation 3**, **Equation 4**, and **Equation 5** of **Paper 1**. Using these equations, and using the notation of **Paper 1**, we obtain the following boundary conditions (requiring continuity of the tangential component of the electric field, or, equivalently, continuity of the scalar potential):

$$\text{BC 1: } -E_c r_1^{t_1} = A_1 r_1 + A_2 / r_1^2,$$

$$\text{BC 2: } A_1 r_2 + A_2 / r_2^2 = -r_2 E_0 + d E_0 / r_2^2$$

Furthermore, we may readily obtain two additional boundary conditions that follow from the requirement that the normal component of the displacement field be continuous at r_1 and r_2 :

$$\mathbf{BC\ 3:} \quad -E_c t_1 r_1^{t_1-1} \varepsilon_r = \left[A_1 - 2A_2/r_1^3 \right] \varepsilon_2$$

$$\mathbf{BC\ 4:} \quad \left[A_1 - 2A_2/r_2^3 \right] \varepsilon_2 = - \left[E_0 + 2dE_0/r_2^3 \right] \varepsilon_3$$

Combining **BC 1** and **BC 3** gives us the following:

$$t_1 \varepsilon_r (\mathbf{BC\ 1}) - r_1 (\mathbf{BC\ 3}) = 0 = A_1 r_1 (t_1 \varepsilon_r - \varepsilon_2) + (A_2/r_1^2) (t_1 \varepsilon_r + 2\varepsilon_2) \rightarrow$$

$$A_2 = A_1 r_1^3 \left[\frac{\varepsilon_2 - t_1 \varepsilon_r}{2\varepsilon_2 + t_1 \varepsilon_r} \right] = A_1 r_1^3 \left(\frac{B_2}{B_3} \right),$$

which, as may be readily checked, is consistent with **Equation 7** of **Paper 1**. At this point, we may conveniently re-express **BC 2** and **BC 4** in terms of only two of the unknowns (A_1 and d) as follows:

$$\mathbf{BC\ 2:} \quad A_1 \left[1 + \frac{\varepsilon_2 - t_1 \varepsilon_r}{2\varepsilon_2 + t_1 \varepsilon_r} \frac{r_1^3}{r_2^3} \right] = E_0 \left[-1 + \frac{d}{r_2^3} \right]$$

$$\mathbf{BC\ 4:} \quad A_1 \left[1 + \frac{2t_1 \varepsilon_r - 2\varepsilon_2}{2\varepsilon_2 + t_1 \varepsilon_r} \frac{r_1^3}{r_2^3} \right] \varepsilon_2 = - \left[1 + 2d/r_2^3 \right] E_0 \varepsilon_3$$

Dividing the second of these two equations by the first (in order to get rid of the unknown quantity, A_1) gives us the following:

$$\varepsilon_3 \left[\frac{2d + r_2^3}{r_2^3 - d} \right] = \varepsilon_2 \left[\frac{2\varepsilon_2(r_2^3 - r_1^3) + t_1 \varepsilon_r (r_2^3 + 2r_1^3)}{\varepsilon_2(2r_2^3 + r_1^3) + t_1 \varepsilon_r (r_2^3 - r_1^3)} \right]$$

At this point, we are almost done (as far as solving for d goes). To proceed, we move all terms proportion to d to the left side of the equation and the terms proportional to r_2^3 to the right side of the equation, giving us the following:

$$d \left[2\varepsilon_3 + \varepsilon_2 \left[\frac{2\varepsilon_2(r_2^3 - r_1^3) + t_1 \varepsilon_r (r_2^3 + 2r_1^3)}{\varepsilon_2(2r_2^3 + r_1^3) + t_1 \varepsilon_r (r_2^3 - r_1^3)} \right] \right] = - \left[\varepsilon_3 - \varepsilon_2 \left[\frac{2\varepsilon_2(r_2^3 - r_1^3) + t_1 \varepsilon_r (r_2^3 + 2r_1^3)}{\varepsilon_2(2r_2^3 + r_1^3) + t_1 \varepsilon_r (r_2^3 - r_1^3)} \right] \right] r_2^3$$

Expressing the above in terms of the quantities P , B_2 and B_3 (defined in **Paper 1**), we obtain the following:

$$\begin{aligned}
d \left[2\varepsilon_3 + \varepsilon_2 \left[\frac{PB_3 - 2B_2}{PB_3 + B_2} \right] \right] &= - \left[\varepsilon_3 - \varepsilon_2 \left[\frac{PB_3 - 2B_2}{PB_3 + B_2} \right] \right] r_2^3 \rightarrow \\
d &= -r_2^3 \left[\frac{\varepsilon_3(PB_3 + B_2) - \varepsilon_2(PB_3 - 2B_2)}{2\varepsilon_3(PB_3 + B_2) + \varepsilon_2(PB_3 - 2B_2)} \right] = -r_2^3 \left[\frac{PB_3(\varepsilon_3 - \varepsilon_2) + B_2(\varepsilon_3 + 2\varepsilon_2)}{PB_3(2\varepsilon_3 + \varepsilon_2) + 2B_2(\varepsilon_3 - \varepsilon_2)} \right] \\
&= -r_2^3 \left[-1 + \frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{PB_3(2\varepsilon_3 + \varepsilon_2) + 2B_2(\varepsilon_3 - \varepsilon_2)} \right] = r_2^3 \left[1 + \frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{B_1 B_2 - PB_3 B_4} \right]
\end{aligned}$$

Thus, the electric field outside of the sphere will be given by (taking the negative gradient of **Equation 5** of **Paper 1**):

$$\begin{aligned}
\left[1 + \frac{2d}{r^3} \right] E_0 \cos(\theta) \mathbf{e}_r + \left[-1 + \frac{d}{r^3} \right] E_0 \sin(\theta) \mathbf{e}_\theta = \\
\left[1 + \frac{2r_2^3}{r^3} \left[1 + \frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{B_1 B_2 - PB_3 B_4} \right] \right] E_0 \cos(\theta) \mathbf{e}_r + \left[-1 + \frac{r_2^3}{r^3} \left[1 + \frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{B_1 B_2 - PB_3 B_4} \right] \right] E_0 \sin(\theta) \mathbf{e}_\theta,
\end{aligned}$$

which is **Equation 8** of **Paper 1**, as desired. We now turn to deriving **Equation 7** of **Paper 1**, which gives the field profile in the gold nanoshell ($r_1 < r < r_2$). From **BC 2**, we already know that A_1 may be readily obtained as follows:

$$\begin{aligned}
A_1 \left[1 + \frac{\varepsilon_2 - t_1 \varepsilon_r}{2\varepsilon_2 + t_1 \varepsilon_r} \frac{r_1^3}{r_2^3} \right] &= E_0 \left[-1 + \frac{d}{r_2^3} \right] = E_0 \left[\frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{B_1 B_2 - PB_3 B_4} \right] \rightarrow \\
A_1 \left[1 + \frac{B_2}{PB_3} \right] &= E_0 \left[\frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{B_1 B_2 - PB_3 B_4} \right] \rightarrow \\
A_1 &= E_0 \left[\frac{3\varepsilon_3 PB_3 + 3\varepsilon_3 B_2}{B_1 B_2 - PB_3 B_4} \right] \left[\frac{PB_3}{PB_3 + B_2} \right] = E_0 \left[\frac{3\varepsilon_3 PB_3}{B_1 B_2 - PB_3 B_4} \right], \\
A_2 &= E_0 r_1^3 \left[\frac{3\varepsilon_3 PB_2}{B_1 B_2 - PB_3 B_4} \right]
\end{aligned}$$

Thus, the electric field in the shell region is given by the following expression:

$$\begin{aligned}
\left[-A_1 + \frac{2A_2}{r^3} \right] \cos(\theta) \mathbf{e}_r + \left[A_1 + \frac{A_2}{r^3} \right] \sin(\theta) \mathbf{e}_\theta = \\
\frac{3\varepsilon_3 P E_0}{B_1 B_2 - PB_3 B_4} \left[\left[-B_3 + 2B_2 \left[\frac{r_1}{r} \right]^3 \right] \cos(\theta) \mathbf{e}_r + \left[B_3 + B_2 \left[\frac{r_1}{r} \right]^3 \right] \sin(\theta) \mathbf{e}_\theta \right],
\end{aligned}$$

which is equivalent to **Equation 7** of **Paper 1**, as desired. Last but not least, we turn to **Equation 6**, which gives the electric field in the radially anisotropic spherical core. From **BC 1**, we have the following:

$$-E_c r_1^{t_1} = A_1 r_1 + A_2 / r_1^2 = \frac{3\varepsilon_3 P E_0 r_1}{B_1 B_2 - P B_3 B_4} [B_3 + B_2] \rightarrow E_c = -\frac{3\varepsilon_3 P E_0 r_1^{1-t_1} (B_3 + B_2)}{B_1 B_2 - P B_3 B_4}$$

From this, we readily obtain the following expression for the electric field for $r < r_1$:

$$\begin{aligned} E_c t_1 r^{t_1-1} \cos(\theta) \mathbf{e}_r - E_c r^{t_1-1} \sin(\theta) \mathbf{e}_\theta &= -\frac{3\varepsilon_3 P E_0 (B_2 + B_3) (r/r_1)^{t_1-1}}{B_1 B_2 - P B_3 B_4} \left[t_1 \cos(\theta) \mathbf{e}_r - \sin(\theta) \mathbf{e}_\theta \right] \\ &= -\frac{9\varepsilon_2 \varepsilon_3 P E_0 (r/r_1)^{t_1-1}}{B_1 B_2 - P B_3 B_4} \left[t_1 \cos(\theta) \mathbf{e}_r - \sin(\theta) \mathbf{e}_\theta \right], \end{aligned}$$

which is **Equation 6** of **Paper 1**. Note that, in simplifying the expression above, we used the fact that $B_2 + B_3 = 3\varepsilon_2$.

Section 3: Maximal field enhancement

In this section, we derive **Equation 13** of **Paper 1**, which corresponds to the maximal field enhancement due to the presence of the nanoshell/core system. First, we note that at $r = r_2 + 0^+$ (just outside of the sphere), the field is given by the following expression (using **Equation 8** of **Paper 1**):

$$\left[3 + \frac{6\varepsilon_3 P B_3 + 6\varepsilon_3 B_2}{B_1 B_2 - P B_3 B_4} \right] E_0 \cos(\theta) \mathbf{e}_r + \frac{3\varepsilon_3 P B_3 + 3\varepsilon_3 B_2}{B_1 B_2 - P B_3 B_4} E_0 \sin(\theta) \mathbf{e}_\theta$$

This is of the form $(3 + 2\alpha) E_0 \cos(\theta) \mathbf{e}_r + \alpha E_0 \sin(\theta) \mathbf{e}_\theta$, which means that we have, in the notation of **Paper 1**,

$$|E_3/E_0|^2 = (3 + 2\alpha)^2 \cos^2(\theta) + \alpha^2 \sin^2(\theta)$$

Taking the derivative of this expression with respect to θ and setting this equal to zero (to find the extrema) then gives us the following expression:

$$-(3 + 2\alpha)^2 \cos(\theta) \sin(\theta) + \alpha^2 \sin(\theta) \cos(\theta) = 0 \rightarrow \sin(2\theta) = 0 \rightarrow \theta = 0, \pi/2, \pi$$

Note that we have another solution given by $\pm\alpha = 3 + 2\alpha \rightarrow \alpha = -1, -3$. Let us briefly examine these edge cases. For these values of α , we have $\mathbf{E}_3 \propto \cos(\theta) \mathbf{e}_r \pm \sin(\theta) \mathbf{e}_\theta$, meaning the field's magnitude is constant everywhere along the boundary. We now return to the non-edge cases. As we saw above, the maximal field magnitude occurs at one or more of $\theta = 0, \pi/2, \pi$. The field magnitude is obviously the same at $\theta = 0$ and $\theta = \pi$. By taking the second derivative of the field magnitude, we see that $\theta = \pi/2$ is a maximum if

$(3 + 2\alpha)^2 - \alpha^2 < 0$ and it is a minimum otherwise. Note that this second derivative condition is just the statement that $\pi/2$ is a maximum if the value of the field magnitude there is larger than that at $\theta = 0, \pi$. The paper seems to assume that the field's maximum is necessarily at $\theta = 0, \pi$; however, I believe this is just a consequence of their choice for material parameters. Note that if $\alpha = -3/2$, for instance, the field maximum will obviously be at $\theta = \pi/2$.

Section 4: Full wave approach

We now go beyond the quasistatic limit and solve for the electromagnetic fields with the effects of retardation included. As we shall see below, closed form solutions, which may be divided into TM and TE channels, may be obtained in the case of radial anisotropy. With an $e^{-i\omega t}$ time dependence, Maxwell's equations read as follows:

$$\nabla \times \mathbf{E}(\mathbf{r}, \omega) = i\omega\mu_0\mathbf{H}(\mathbf{r}, \omega), \nabla \times \mathbf{H}(\mathbf{r}, \omega) = -i\omega\varepsilon_0\varepsilon\mathbf{E}(\mathbf{r}, \omega)$$

We are considering non-magnetic systems (i.e. $\mu_{ij} = \delta_{ij}$) with uniaxial (in the spherical coordinate basis) permittivities (as in the last section).

For TM modes, we take $\mathbf{H}(\mathbf{r}, \omega) = \mathbf{r} \times \nabla [f(r, \omega)Y_{lm}(\theta, \phi)]$, where $f(r, \omega)$ depends on the particulars of the permittivity tensor. Similarly, for TE modes, we take

$$\mathbf{E}(\mathbf{r}, \omega) = \mathbf{r} \times \nabla [f(r, \omega)Y_{lm}(\theta, \phi)]. \text{ Note that we can equivalently write}$$

$\mathbf{D}(\mathbf{r}, \omega) = \mathbf{r} \times \nabla (\varepsilon_t f(r, \omega)Y_{lm}(\theta, \phi))$, where ε_t is the permittivity along the $\mathbf{e}_\theta$ and $\mathbf{e}_\phi$ directions (see, for instance, **Equation 1** of **Paper 2**). This is because the electric field for TE modes is purely tangential to the spherical boundary.

We first consider the TM channel, for which we obtain the following (using

$$\mathbf{e}_r \times \mathbf{e}_\phi = -\mathbf{e}_\theta, \mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{e}_\phi):$$

$$\mathbf{H}(\mathbf{r}, \omega) = f(r, \omega) \left[-\frac{1}{\sin(\theta)} \partial_\phi Y_{lm}(\theta, \phi) \mathbf{e}_\theta + \partial_\theta Y_{lm}(\theta, \phi) \mathbf{e}_\phi \right]$$

From this expression, we may readily use the Maxwell-Ampere law to obtain the following expression for the electric field:

$$\begin{aligned} \mathbf{E}(\mathbf{r}, \omega) = & \frac{i}{\omega r \varepsilon_0 \varepsilon_t} \left[\frac{\varepsilon_t}{\varepsilon_r} \left[\frac{1}{\sin(\theta)} \partial_\theta [\sin(\theta) \partial_\theta Y_{lm}(\theta, \phi)] + \frac{1}{\sin^2(\theta)} \partial_\phi^2 Y_{lm}(\theta, \phi) \right] f(r, \omega) \mathbf{e}_r + \right. \\ & \left. - \left[\partial_\theta Y_{lm}(\theta, \phi) \mathbf{e}_\theta + \frac{1}{\sin(\theta)} \partial_\phi Y_{lm}(\theta, \phi) \mathbf{e}_\phi \right] \partial_r (r f(r, \omega)) \right] \end{aligned}$$

Perhaps it is obvious, but we note that the radial component of the above has a factor of ε_r^{-1} and the tangential components have a factor of ε_t^{-1} . At this point, we also note that the radial component of the electric field may be significantly simplified by using the fact that

the spherical harmonics are eigenfunctions of the angular momentum operator squared: i.e., $r^2 \nabla^2 Y_{lm}(\theta, \phi) = -l(l+1)Y_{lm}(\theta, \phi)$. Thus, we may re-write the radial part of the electric field as follows:

$$\mathbf{e}_r \cdot \mathbf{E}(\mathbf{r}, \omega) = -\frac{i}{\omega r \epsilon_0 \epsilon_t} \left[\frac{\epsilon_t}{\epsilon_r} l(l+1) f(r, \omega) Y_{lm}(\theta, \phi) \mathbf{e}_r \right]$$

Using Faraday's law, we then obtain the following equation for $f(r, \omega)$:

$$-\frac{i\omega\mu_0}{\sin(\theta)} f(r, \omega) \partial_\phi Y_{lm}(\theta, \phi) = \frac{1}{r} \left[-\frac{il(l+1)}{\omega r \epsilon_0 \epsilon_r \sin(\theta)} f(r, \omega) + \frac{i}{\omega \epsilon_0 \epsilon_t \sin(\theta)} \partial_r^2 (r f(r, \omega)) \right] \times \partial_\phi Y_{lm}(\theta, \phi)$$

Note that the equation above was derived by equating $i\omega\mu_0 \mathbf{H}(\mathbf{r}, \omega) \cdot \mathbf{e}_\theta$ with $\mathbf{e}_\theta \cdot [\nabla \times \mathbf{E}(\mathbf{r}, \omega)]$. Of course, the same equation may be derived by equating

$i\omega\mu_0 \mathbf{H}(\mathbf{r}, \omega) \cdot \mathbf{e}_\phi$ with $\mathbf{e}_\phi \cdot [\nabla \times \mathbf{E}(\mathbf{r}, \omega)]$, which we do below:

$$i\omega\mu_0 f(r, \omega) \partial_\theta Y_{lm}(\theta, \phi) = \frac{i}{r\omega\epsilon_0} \left[-\frac{1}{\epsilon_t} \partial_r^2 (r f(r, \omega)) + \frac{l(l+1)}{\epsilon_r r} f(r, \omega) \right] \partial_\theta Y_{lm}(\theta, \phi)$$

Thus, in either case, our equation for $f(r, \omega)$ is given by the following differential equation:

$$-r^2 \epsilon_t (\omega/c)^2 f(r, \omega) = -\frac{l(l+1)\epsilon_t}{\epsilon_r} f(r, \omega) + r \partial_r^2 (r f(r, \omega)) \rightarrow$$

$$r^2 f''(r, \omega) + 2r f'(r, \omega) + \left[r^2 \epsilon_t (\omega/c)^2 - l(l+1)(\epsilon_t/\epsilon_r) \right] f(r, \omega) = 0$$

Note that by defining $g(r, \omega) \equiv r f(r, \omega)$, we may equivalently write the above as follows:

$$\partial_r^2 g(r, \omega) + \left[\epsilon_t (\omega/c)^2 - \frac{l(l+1)(\epsilon_t/\epsilon_r)}{r^2} \right] g(r, \omega) = 0$$

Thus, note that $f(r, \omega) = \{j/y\}_{l'}(r\omega\sqrt{\epsilon_t}/c)$, where $j_{l'}$ is a spherical Bessel function of the first kind, $y_{l'}$ is a spherical Bessel function of the second kind and l' satisfies the following:

$$l'(l'+1) = l(l+1)\epsilon_t/\epsilon_r$$

From **Section 1**, we already know that the solution to this quadratic equation is given by the following:

$$l' = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{l(l+1)\epsilon_t(\omega)}{\epsilon_r(\omega)}}$$

Section 5: Surface scattering

In this section, we do a somewhat novel calculation of the effect of surface scattering in a nanosphere of radius R . The most basic approximation to the geometric mean free path, l , in a nanosphere is given by an angle-averaged evaluation of the mean free path (assuming isotropic scattering):

$$\begin{aligned}
l &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{\pi/2} \sqrt{2R^2 - 2R^2 \cos(\pi - 2\theta)} \sin(\theta) d\theta d\phi = \frac{2R}{2\pi} \int_0^{2\pi} \int_0^{\pi/2} \sin(\theta) \cos(\theta) d\theta d\phi \\
&= R \int_0^{\pi/2} \sin(2\theta) d\theta = -(R/2)(-2) = R,
\end{aligned}$$

where the factor of $(2\pi)^{-1}$ is a division by the range of solid angles in the upper hemisphere. A more sophisticated approach (which I am not sure has been done before) is given by an angle-averaged **transport-limited** mean free path (see **Equation 3.1** of **Paper 12**), given by the following:

$$\begin{aligned}
l &= \int_0^{\pi/2} \lambda_0 \left[1 - \exp \left[-2 \cos(\theta) R / \lambda_0 \right] \right] \sin(\theta) d\theta = \lambda_0 - \lambda_0 \int_{-1}^0 \exp \left[2uR / \lambda_0 \right] du \\
&= \lambda_0 - (\lambda_0^2 / 2R) \left[1 - \exp \left[-2R / \lambda_0 \right] \right],
\end{aligned}$$

where $\lambda_0 = v_F \tau$ is the mean free path of the bulk material. As a sanity check, we note that for $R / \lambda_0 \ll 1$ we obtain:

$$l \approx \lambda_0 - (\lambda_0^2 / 2R) \left[1 - \left[1 - 2R / \lambda_0 + 2R^2 / \lambda_0^2 + O((2R / \lambda_0)^3) \right] \right] \approx R$$

Thus, if the mean free path of the underlying bulk material is much larger than the radius of the sphere (and thus larger than any chord length in the sphere), we recover our original, more naive result, $l \approx R$, as expected.

Section 6: Surface scattering in a rectangular wire

In this section, we formulate the effect of surface scattering in a rectangular wire, following the supplemental information of **Paper 8**. To start, we re-state the Boltzmann transport equation in the DC limit (using the notation of **Paper 8**):

$$v_y(\mathbf{p}) \partial_y \delta f(\mathbf{p}, y, z) + v_z(\mathbf{p}) \partial_z \delta f(\mathbf{p}, y, z) + (\delta f(\mathbf{p}, y, z) / \tau) = -\dot{f}_0(\mathbf{p})$$

$$\text{where } \dot{f}_0 = -e \mathbf{E} \cdot \nabla_{\mathbf{p}} f_0(\varepsilon(\mathbf{p})) = -e E_x f'_0(\varepsilon_{\mathbf{p}}) v_x(\mathbf{p}),$$

where we have taken the electric field to be constant and polarized along the direction of the wire. To solve this equation, we define an auxiliary function, $g(s)$, given as follows:

$$g_{\mathbf{p}}(s) \equiv \delta f(y_0 + v_y(\mathbf{p})s, z_0 + v_z(\mathbf{p})s),$$

where (y_0, z_0) lies on the appropriate boundary for which $\delta f(\mathbf{p}, y_0, z_0) = 0$ (given by the diffusive boundary condition). Thus, we can now write an ordinary differential equation for our auxiliary function given by the following:

$$\begin{aligned}
\frac{dg_{\mathbf{p}}(s)}{ds} + g_{\mathbf{p}}(s) / \tau &= e^{-s/\tau} \frac{d}{ds} \left[g_{\mathbf{p}}(s) e^{s/\tau} \right] = -\dot{f}_0(\mathbf{p}) \rightarrow \\
g_{\mathbf{p}}(s) &= -\tau \dot{f}_0(\mathbf{p}) e^{-s/\tau} \left[e^{s/\tau} - 1 \right] = -\tau \dot{f}_0(\mathbf{p}) \left[1 - e^{-r/v_t \tau} \right],
\end{aligned}$$

where

$$r = s\sqrt{v_y^2 + v_z^2} \equiv sv_t$$

Note that the above is **Equation S23 of Paper 8** (their equation has a typo in the argument of the exponent). We now wish to integrate the contribution of δf to the current.

$$\int \int \delta f(\mathbf{p}, y, z) dy dz = \left[v_z(\mathbf{p})/v_t(\mathbf{p}) \right] \int \int \delta f(\mathbf{p}, u + \left[v_y(\mathbf{p})/v_t(\mathbf{p}) \right] r, \left[v_z(\mathbf{p})/v_t(\mathbf{p}) \right] r) du dr,$$

where we have performed the following change of variables:

$$y(u, r) = u + \left[v_y(\mathbf{p})/v_t(\mathbf{p}) \right] r, z(u, r) = \left[v_z(\mathbf{p})/v_t(\mathbf{p}) \right] r,$$

where u will range from 0 to w (the width of the rectangular nanowire) and r will range from 0 to the maximal chord length inside the rectangular wire. We first perform the integral over r . The contribution at fixed u of the chord going at angle $\theta = \tan^{-1}(v_z(\mathbf{p})/v_y(\mathbf{p}))$ is given by the following expression:

$$\int \delta f(\mathbf{p}, u + \left[v_y(\mathbf{p})/v_t(\mathbf{p}) \right] r, \left[v_z(\mathbf{p})/v_t(\mathbf{p}) \right] r) dr = -\tau \dot{f}_0(\mathbf{p}) \left[l(u) + v_t \tau \left[e^{-l(u)/v_t \tau} - 1 \right] \right] = -\tau \dot{f}_0 \left[l(u) - v_t \tau \left[1 - e^{-l(u)/v_t \tau} \right] \right],$$

which is **Equation S24 of Paper 8**. Note that $l(u)$ is the maximal chord length. At this point, we just need to find this maximal chord length to evaluate the remaining integral. We do this in the following subsection.

Subsection 6.1: Derivation of $l(u)$

To stay consistent with the notation of **Paper 8**, we use $l(y)$ instead of $l(u)$ in the following. Let's say we start on the edge located at $z = 0$ (i.e. our starting location is $(y, z) = (y, 0)$). We traverse the line given by $(y + v_y s, z + v_z s)$ until we hit one of the edges. If we hit the top edge, i.e. the line located at $z = h$, we will have $z + v_z s = h$. If we hit the edge on the right, i.e. the line located at $y = w$, we will have $y + v_y s = w$. Therefore,

$$0 < s < \min\{h/v_z, (w - y)/v_y\} \leftrightarrow 0 < r < \min\{h\sqrt{1 + (v_y/v_z)^2}, (w - y)\sqrt{1 + (v_z/v_y)^2}\}$$

At this point, it is obvious that the maximal chord length depends on the sign of $w - h(v_y/v_z)$. If $w - h(v_y/v_z) > 0 \rightarrow w > hv_y/v_z$, we will have to split the integral up into two parts: one for $y < w - hv_y/v_z$, for which $l(y) = hv_t/v_z$ and the remaining region for which $l(y) = (w - y)v_t/v_y$. On the other hand, if $w < hv_y/v_z$, we will have

$l(y) = (w - y)v_t/v_y$ for the whole region of integration. For $w < hv_y/v_z$, we obtain:

$$l(y) = \frac{v_z}{v_t} (-\tau \dot{f}_0) \Theta(hv_y/v_z - w) \left[\int_0^w \left[\frac{(w - y)v_t}{v_y} - v_t \tau + v_t \tau \exp \left[- (w - y)/(v_y \tau) \right] \right] dy \right] \\ = \frac{v_z}{v_t} (-\tau \dot{f}_0) \Theta(hv_y/v_z - w) \left[\frac{w^2 v_t}{2 v_y} - v_t \tau w + \tau^2 v_t v_y \left[1 - \exp \left[- w/(v_y \tau) \right] \right] \right],$$

where $\Theta(x)$ is the Heaviside step function. This is consistent with the formula given on

Page S5 after **Equation S25** of **Paper 8**. We now calculate for the other possibility (

$hv_y < wv_z$):

$$l(y) = \frac{v_z}{v_t} (-\tau \dot{f}_0) \Theta(w - hv_y/v_z) \left[-v_t \tau w + \left[w - hv_y/v_z \right] \left[(hv_t/v_z) + v_t \tau \exp \left[-h/(v_z \tau) \right] \right] + \int_{w-hv_y/v_z}^w \left[(w-y)v_t/v_y + v_t \tau \exp \left[-(w-y)/(v_y \tau) \right] \right] dy \right]$$

To proceed, we evaluate the integral, which gives us the following expression:

$$hw(v_y/v_z)(v_t/v_y) - (v_t/v_y) \left[hw(v_y/v_z) - (h^2/2)(v_y/v_z)^2 \right] + (v_t \tau)(v_y \tau) \left[1 - \exp \left[-h/(v_z \tau) \right] \right]$$

Thus, the total answer (for $w > hv_y/v_z$) is given by the following:

$$l(y) = -\tau \dot{f}_0 \Theta(w - hv_y/v_z) \left[-v_z \tau w + hw - (h^2/2)(v_y/v_z) + v_z \tau \left[w - hv_y/v_z - v_y \tau \right] \exp \left[-h/(v_z \tau) \right] + (v_z \tau)(v_y \tau) \right],$$

which is also consistent with **Page S5** of **Paper 8**. Note that our derivation only dealt with chords intersecting the bottom boundary. Analogous calculations may be done for the other boundaries.

Section 7: Boltzmann transport equation in a metallic sphere

In this section, we solve the Boltzmann transport equation in a metallic sphere of radius R . We put the sphere in a spatially uniform continuous wave electric field, $\mathbf{E}(\omega)e^{-i\omega t}$ (which is a valid approximation for small spheres). We operate in the frequency domain, meaning the first order change in the electronic occupations is given by $f^{(1)}(\mathbf{p}, \mathbf{r})e^{-i\omega t}$. The Boltzmann transport equation for $f^{(1)}(\mathbf{p}, \mathbf{r})$ is thus given by the following:

$$\mathbf{v}(\mathbf{p}) \cdot \nabla_{\mathbf{r}} f^{(1)}(\mathbf{p}, \mathbf{r}) - e\mathbf{E}(\omega) \cdot \nabla_{\mathbf{p}} f^{(0)}(\varepsilon(\mathbf{p})) - i\omega f^{(1)}(\mathbf{p}, \mathbf{r}) = -f^{(1)}(\mathbf{p}, \mathbf{r})/\tau \rightarrow$$

$$\mathbf{v}(\mathbf{p}) \cdot \nabla_{\mathbf{r}} f^{(1)}(\mathbf{p}, \mathbf{r}) - e\mathbf{E}(\omega) \cdot \nabla_{\mathbf{p}} f^{(0)}(\varepsilon(\mathbf{p})) = i \left[\omega + i/\tau \right] f^{(1)}(\mathbf{p}, \mathbf{r})$$

Once $f^{(1)}(\mathbf{p}, \mathbf{r})$ is obtained for all momenta, $\mathbf{p}$, the current density at any coordinate, $\mathbf{r}$, may be easily obtained through the following expression:

$$\mathbf{J}(\mathbf{r}, \omega) = \frac{(-e)g_s}{(2\pi\hbar)^3} \int \mathbf{v}(\mathbf{p}) f^{(1)}(\mathbf{p}, \mathbf{r}) d^3\mathbf{p},$$

where $g_s = 2$ is the spin degeneracy and the prefactor of $(2\pi\hbar)^{-3}$ arises from the quantization of wavenumbers in all three directions. Let's consider a generic point within the sphere, $\mathbf{r}$. In order to use the results of the previous section, we wish to find the distance a particle, with velocity $\mathbf{v}(\mathbf{p})$, must travel in order to reach this point from the boundary of the sphere. This is trivially calculated as follows:

$$|\mathbf{r} - \mathbf{v}(\mathbf{p})s|^2 = R^2 \rightarrow -2s(\mathbf{v}(\mathbf{p}) \cdot \mathbf{r}) + v_F^2 s^2 = R^2 - r^2 \rightarrow$$

$$(\mathbf{v}(\mathbf{p}) \cdot \mathbf{r}) \pm \sqrt{(\mathbf{v}(\mathbf{p}) \cdot \mathbf{r})^2 + v_F^2(R^2 - r^2)}$$

$v_F s = \frac{(\mathbf{v}(\mathbf{p}) \cdot \mathbf{r}) \pm \sqrt{(\mathbf{v}(\mathbf{p}) \cdot \mathbf{r})^2 + v_F^2(R^2 - r^2)}}{v_F^2}$, where we used the fact that $|\mathbf{v}(\mathbf{p})|^2 = v_F^2$ for any $\mathbf{p}$ at the Fermi surface.

Note that the discriminant in the expression above is greater than zero:

$$(\mathbf{v}(\mathbf{p}) \cdot \mathbf{r})^2 - v_F^2 r^2 + v_F^2 R^2 = v_F^2 [R^2 - r^2 \sin^2(\theta_{\mathbf{p}, \mathbf{r}})] > 0 \text{ where } \theta_{\mathbf{p}, \mathbf{r}} \equiv \mathbf{p} \cdot \mathbf{r} / (|\mathbf{p}||\mathbf{r}|),$$

as obviously required. At this point, we must determine the correct root of s . First, we note that for any $\mathbf{r}$ and for any $\mathbf{p}$, one root of s will be positive and the other will be negative.

The "correct" root of s will be the positive one, since this corresponds to the electron leaving the sphere's boundary with velocity $\mathbf{v}(\mathbf{p})$ and ending up at $\mathbf{r}$. With this established, we may use the results of the previous section to find the first order change in the electronic distribution at $\mathbf{r}$:

$$f^{(1)}(\mathbf{p}, \mathbf{r}) = - \left[\frac{1}{1/\tau - i\omega} \right] \dot{f}_0(\mathbf{p}) \left[1 - e^{-(1/\tau - i\omega)l(\mathbf{r})/v_F} \right]$$

$$= \left[(-e) \frac{\delta(\varepsilon(\mathbf{p}) - \varepsilon_F)(\mathbf{E}(\omega) \cdot \mathbf{v}(\mathbf{p}))}{1/\tau - i\omega} \right] \left[1 - e^{-(1/\tau - i\omega)l(\mathbf{r})/v_F} \right],$$

where $l(\mathbf{r}) \equiv r \cos(\theta) + \sqrt{R^2 - r^2 \sin^2(\theta)}$ and we have dropped the subscripts on θ

This is consistent with **Equation 38** of **Paper 11**, **Equation 22** of **Paper 9** and **Equation 15** of **Paper 10**. At this point, we wish to calculate the average current response in the nanoshell.

This corresponds to the following expression:

$$\langle \mathbf{J}(\mathbf{r}, \omega) \rangle \equiv \frac{3}{4\pi R^3} \int \mathbf{J}(\mathbf{r}, \omega) d^3 \mathbf{r} =$$

$$\frac{g_s e}{(2\pi \hbar)^3} \int \mathbf{v}(\mathbf{p}) \left[\frac{3(2\pi)}{4\pi R^3} \left[\frac{1}{1/\tau - i\omega} \right] \dot{f}_0(\mathbf{p}) \right] \int_0^R \int_{-1}^1 \left[1 - \exp \alpha \left[r \cos(\theta) + \sqrt{R^2 - r^2 \sin^2(\theta)} \right] \right] r^2 d \cos(\theta) dr d^3 \mathbf{p},$$

where the additional factor of (2π) comes from integration over the azimuthal direction, and we have, for convenience, defined α as follows:

$$\alpha \equiv - \left[\frac{1}{\tau} - i\omega \right] \frac{1}{v_F}$$

In order to compute the integral over the spatial directions, we make a judicious u-substitution (taking the argument of the exponential to be the "u" of choice). This gives us the following:

$$u(\theta) = \alpha \left[r \cos(\theta) + \sqrt{R^2 - r^2 \sin^2(\theta)} \right] \rightarrow \frac{du(\theta)}{d\theta} = \alpha \left[-r \sin(\theta) - \frac{r^2 \sin(\theta) \cos(\theta)}{\sqrt{R^2 - r^2 \sin^2(\theta)}} \right] =$$

$$- \alpha r \sin(\theta) \left[1 + \frac{r \cos(\theta)}{\sqrt{R^2 - r^2 \sin^2(\theta)}} \right] = \frac{-r \sin(\theta) u(\theta)}{\sqrt{R^2 - r^2 \sin^2(\theta)}}$$

This is now almost written in a useful form. In order to perform the spatial integrals, we need now just write the denominator of this expression in terms of $u(\theta)$. This is trivially accomplished as follows:

$$\frac{1}{u(\theta)} = \frac{1}{\alpha} \left[r \cos(\theta) - \sqrt{R^2 - r^2 \sin^2(\theta)} \right] \frac{1}{r^2 - R^2} \rightarrow$$

$$\sqrt{R^2 - r^2 \sin^2(\theta)} = \frac{1}{2} \left[\frac{u(\theta)}{\alpha} - \frac{(r^2 - R^2)\alpha}{u(\theta)} \right]$$

Thus, we may now easily re-express the integration measure (over the polar angle, θ) as follows:

$$\sin(\theta) d\theta = -\frac{1}{2ru(\theta)} \left[\frac{u(\theta)}{\alpha} - \frac{(r^2 - R^2)\alpha}{u(\theta)} \right] du$$

Thus, we re-express our integral over the spatial coordinates as follows:

$$2\pi \int_0^R \int_{-1}^1 \left[1 - \exp(u(\theta)) \right] d\cos(\theta) r^2 dr = \frac{4}{3} \pi R^3 - \pi \int_0^R \int_{\alpha(R-r)}^{\alpha(R+r)} \exp(u) \left[\frac{1}{\alpha} + \frac{(R^2 - r^2)\alpha}{u^2} \right] du dr$$

To evaluate this integral, it is convenient to swap the order of integration as follows:

$$\int_0^R \int_{\alpha(R-r)}^{\alpha(R+r)} f(u, r) du dr = \int_0^R \int_0^{2\alpha R} \Theta(u - \alpha(R - r)) \Theta(\alpha(R + r) - u) f(u, r) du dr =$$

$$\int_0^R \int_0^{2\alpha R} \Theta(r - R + u/\alpha) \Theta(r + R - u/\alpha) f(u, r) du dr = \int_0^{2\alpha R} \int_{|R-u/\alpha|}^R f(u, r) du dr$$

The above holds for a generic function of two variables, $f(u, r)$. Specializing to our case of interest we find:

$$\begin{aligned}
& \frac{\pi}{\alpha} \int_0^R \int_{\alpha(R-r)}^{\alpha(R+r)} \exp(u) \left[1 + \frac{(R^2 - r^2)\alpha^2}{u^2} \right] du dr \\
&= \frac{\pi}{\alpha} \int_0^{2\alpha R} \int_{|R-u/\alpha|}^R \exp(u) \left[1 + \frac{\alpha^2(R^2 - r^2)}{u^2} \right] r dr du \\
&= \frac{\pi}{\alpha} \int_0^{2\alpha R} \exp(u) \left[(R^2/2)(1 + \alpha^2 R^2/u^2) - \alpha^2(R^4/4u^2) + \right. \\
&\quad \left. - ((R - u/\alpha)^2/2)(1 + \alpha^2 R^2/u^2) + \alpha^2(R - u/\alpha)^4/(4u^2) \right] du \\
&= \frac{\pi}{\alpha} \int_0^{2\alpha R} \exp(u) \left[(R^2/2) - (-Ru/\alpha + u^2/(2\alpha^2))(1 + \alpha^2 R^2/u^2) - (R^2/2) + \right. \\
&\quad \left. \alpha^2/(4u^2) \left[-4R^3(u/\alpha) + 6R^2(u/\alpha)^2 - 4R(u/\alpha)^3 + (u/\alpha)^4 \right] \right] du \\
&= \frac{\pi}{\alpha} \int_0^{2\alpha R} \exp(u) \left[-u^2/(4\alpha^2) + R^2 \right] du
\end{aligned}$$

We now note that the integrals over u may be trivially integrated by parts as follows:

$$\begin{aligned}
\int \exp(u) u du &= u \exp u - \exp(u), \\
\int \exp(u) u^2 du &= u^2 \exp(u) - 2 \int \exp(u) u du = \exp(u) [u^2 - 2u + 2]
\end{aligned}$$

Thus, the integral is given by the following:

$$\begin{aligned}
& \frac{\pi}{\alpha} \int_0^{2\alpha R} \exp u \left[R^2 - u^2/(4\alpha^2) \right] du = \frac{\pi}{\alpha} \left[\exp(u) \left[R^2 - \frac{(u^2 - 2u + 2)}{4\alpha^2} \right] \right]_0^{2\alpha R} \\
&= -\pi \exp(2\alpha R)/(4\alpha^3) \left[(2\alpha R)^2 - 4\alpha R + 2 - 4\alpha^2 R^2 \right] + \pi/(2\alpha^3) - \pi R^2/\alpha \\
&= -\pi \exp(2\alpha R)/(2\alpha^3) \left[1 - 2\alpha R \right] + \pi/(2\alpha^3) - \pi R^2/\alpha
\end{aligned}$$

Thus, we readily obtain the following final answer for our integral over the spatial variables:

$$\begin{aligned}
& 2\pi \int_0^R \int_{-1}^1 \left[1 - \exp(u(\theta)) \right] d \cos(\theta) r^2 dr = \\
& \pi R^3 \left[\frac{4}{3} + \frac{1}{(\alpha R)} - \frac{1}{2(\alpha R)^3} + \frac{\exp(2\alpha R)}{2(\alpha R)^3} \left[1 - 2(\alpha R) \right] \right]
\end{aligned}$$

In order to make a connection to the existing literature, we note that our constant, α , is related to the constant, q , defined by **Equation 22** of **Paper 10** and **Equation 27** of **Paper 9**.

In particular, $\alpha R = -q/2$, which means that we may write the above as:

$$\pi R^3 \left[\frac{4}{3} - \frac{2}{q} + \frac{4}{q^3} - \frac{4 \exp(-q)}{q^3} \left[1 + q \right] \right]$$

The quantity in brackets may be recognized as the function $\Psi(q)$ defined by **Equation 21** of **Paper 10** or **Equation 27** of **Paper 9**. Moving on, we may now evaluate the average current response, which is given by the following:

$$\langle \mathbf{J}(\mathbf{r}, \omega) \rangle = \frac{2e^2 E}{8\pi^3 \hbar^3} \left[\frac{3}{4} \right] \int \left[\frac{4}{3} + \frac{1}{(\alpha R)} - \frac{1}{2(\alpha R)^3} + \frac{\exp(2\alpha R)}{2(\alpha R)^3} \left[1 - 2(\alpha R) \right] \right] \times$$

$$\left[\frac{1}{1/\tau - i\omega} \right] (v_F^2 \cos^2(\theta)) \sin(\theta) \delta(p - mv_F) \left[v_F^{-1} \right] (mv_F)^2 (2\pi) dp,$$

where we used the following expression(s) for the derivative of the Fermi-Dirac distribution (noting that the Fermi energy, ε_F is related to the Fermi momentum, p_F and the Fermi velocity, v_F through $\varepsilon_F = p_F^2/2m = mv_F^2/2$):

$$\partial_\varepsilon f^{(0)}(\varepsilon) \approx -\delta(\varepsilon - \varepsilon_F) = \delta(p - p_F)/v_F = \delta(p - mv_F)/v_F$$

Thus, the average conductivity is given by the following:

$$\frac{3e^2}{4\pi^2 \hbar^3} \left[\frac{1}{1/\tau - i\omega} \right] \left[\int_0^{\pi/2} \cos^2(\theta) \sin(\theta) d\theta \right] m^2 v_F^3 \times$$

$$\left[\frac{4}{3} + \frac{1}{(\alpha R)} - \frac{1}{2(\alpha R)^3} + \frac{\exp(2\alpha R)}{2(\alpha R)^3} \left[1 - 2(\alpha R) \right] \right] =$$

$$\frac{9n_0 e^2}{4m} \left[\frac{1}{1/\tau - i\omega} \right] \left[\int_0^{\pi/2} \cos^2(\theta) \sin(\theta) d\theta \right] \left[\frac{4}{3} + \frac{1}{(\alpha R)} - \frac{1}{2(\alpha R)^3} + \frac{\exp(2\alpha R)}{2(\alpha R)^3} \left[1 - 2(\alpha R) \right] \right]$$

where, just to make sure we are all on the same page, we iterate/reiterate that n_0 is the electronic density and m is the electronic effective mass. Note that the above is consistent with **Equation 34** and **Equation 35** of **Paper 10**. At this point, we note that the angular variable is trivial:

$$\int_0^{\pi/2} \sin(\theta) \cos^2(\theta) d\theta = \int_0^1 u^2 du = 1/3$$

Thus, the conductivity of the nanosphere, $\sigma(\omega)$, may be expressed as follows:

$$\sigma(\omega) = \frac{3n_0 e^2}{4m} \left[\frac{1}{1/\tau - i\omega} \right] \left[\frac{4}{3} + \frac{1}{(\alpha R)} - \frac{1}{2(\alpha R)^3} + \frac{\exp(2\alpha R)}{2(\alpha R)^3} \left[1 - 2(\alpha R) \right] \right],$$

Note that in the limit of large R , we obtain the bulk conductivity (as expected):

$$\sigma(\omega) \rightarrow \frac{in_0 e^2}{m(\omega + i/\tau)}$$